Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
C		2627351

Behind the Ballroom Scores: Reconstructing Fan Voting and Designing Fairer Rules for DWTS

Summary

In the digital entertainment era, televised dance competitions blend professional judging with audience participation, yet the audience vote totals are usually hidden from viewers and contestants. This lack of transparency makes it difficult to understand whether weekly eliminations reflect dance quality, popularity, or both. To address this issue, we analyzed season-level and week-level competition data and built a modeling framework that infers unseen fan support and evaluates how different voting rules shape outcomes.

Firstly, we infer hidden fan support using a two part strategy. **Feasible-Set Model** quantifies uncertainty via elimination constraints, and **Likelihood Model** yields stable point reconstructions. Internal diagnostics show strong feasibility and consistency. Model A is feasible in **96.6 percent** of weeks and its representative point matches the observed elimination in **98.8 percent** of feasible weeks, while Model B matches observed eliminations in **97.7 percent** of weeks. A one week ahead sanity check over **213** eligible single-elimination weeks (excluding no/multi-elimination or withdrawal weeks) yields hit rates of **98.59 percent** for Percentage and **78.87 percent** for Rank. Uncertainty is quantified and remains nontrivial, with average interval width **0.1267** and normalized entropy mean **0.9344**.

Next, we conduct **weekly counterfactual simulations** to compare two common aggregation rules, **Rank** and **Percentage**. Using inferred fan share estimates and confirming via a cross inference audit, Percentage shows stronger fan alignment, with mean ρ_{fan} near 0.78 versus about 0.70 under Rank. The rules disagree in **18.94%** of analysis weeks (single-elimination weeks after cleaning), and in those conflicts Percentage empirically retains the higher-fan-share contestant, indicating that Percentage structurally amplifies fan influence when judge scores are close and fan support is uneven.

Thirdly, we fit a **mixed-effects model** to separate stable pro and contestant differences from week-to-week noise. In-sample fit reaches **RMSE 0.6075** for judges and **0.5740** for fans, and conditional R^2 remains substantially larger than marginal R^2 for both outcomes, showing that persistent random effects explain meaningful variance beyond observed covariates. This decomposition helps distinguish performance-related signals from long-run popularity or partnership advantages.

Finally, we introduce **MARS-Soft+Disagreement**, a revised weekly elimination mechanism that adds limited protection when strong judging support conflicts with unusually low inferred fan support. Tuning on early seasons selects $\alpha = 0.20$**,** $\beta = 0.20$**,** $p = 2$. On held-out seasons, the rule achieves **RescueRate 0.789** while keeping disruption moderate with **ChangeRate 0.385** and maintaining a small balance gap with $\Delta\rho = 0.210$. With a nontrivial **FII 0.161**, the results suggest improved fairness and transparency without removing the meaningful role of audience participation.

Keywords: Fan vote inference, Elimination simulation, Voting rule comparison, Hierarchical effects, Fairness oriented design

Contents

1 Introduction **3**
1.1 Problem Background . . . 3
1.2 Restatement of the Problem . . . 3
1.3 Data Cleaning . . . 3
1.4 Our Work . . . 4

2 Assumptions **4**

3 Notations **5**

4 Fan Vote Inference via Feasible-Set and Likelihood Models **6**
4.1 Model Overview and Construction . . . 6
4.2 Diagnostic Plot for Judge Signal and Fan Share . . . 7
4.3 Validation and Uncertainty Quantification . . . 8

5 Voting Mechanism Audit via Counterfactual Analysis **10**
5.1 Week-by-Week Counterfactual Simulator . . . 10
5.2 Key Evidence of Fan Amplification . . . 11
5.3 Why Percentage Flips Eliminations: "Variance Engulfing" . . . 12
5.4 Controversial Figures: What Changes Under Alternative Rules? . . . 13
5.5 Policy Recommendation . . . 13

6 Impact of Pros and Celebrity Characteristics: Mixed-Effects Model **13**
6.1 Data unit & outcomes . . . 13
6.2 Model specification . . . 14
6.3 Mixed effects results and interpretation . . . 14
6.4 Robustness checks . . . 17

7 Proposed Weekly Voting Rule: MARS-Soft+Disagreement **17**
7.1 Design Rationale and Mechanism Overview . . . 17
7.2 Rule definition . . . 18
7.3 Evaluation protocol and metrics . . . 19
7.4 Results and discussion . . . 19

8 Model Sensitivity Analysis **20**
8.1 Fan-Vote Inference . . . 20
8.2 Pros & Traits Impact . . . 20

9 Overall Model Evaluation **21**
9.1 Strengths 21
9.2 Weaknesses 21

References **22**

Memorandum **23**

Appendices **25**

1 Introduction

1.1 Problem Background

Dancing with the Stars is a long-running competition in which judges assign performance scores and viewers cast votes that jointly determine weekly eliminations and the season champion. Because fan vote totals are not publicly released, it is difficult for the audience to understand or verify why certain contestants advance, especially when popularity and technical quality point to different outcomes. Past disputes and repeated rule adjustments in how scores and votes are combined show that small changes in the aggregation method can substantially alter eliminations and perceived fairness. Using the official season-by-week records in the 2026 MCM Problem C dataset, we infer hidden fan-vote signals, compare alternative combining rules, and design a fairer and more transparent system that still preserves audience engagement [1, 2].

1.2 Restatement of the Problem

We need to analyze the data provided by DWTS and answer the following questions:

- **Task 1: Estimate Weekly Fan Vote Shares**

  Build a model to estimate each contestant's weekly *fan vote share* and a *scale free* popularity signal. Check whether the implied rankings align with eliminations. Quantify uncertainty across contestants and weeks.

- **Task2: Evaluate Two Official Mechanisms**

  Combine estimated *fan vote shares* with judges' scores to simulate both official scoring methods for each season. Compare eliminations, final rankings, and hand advantages. Study controversial cases and evaluate adding a judges' bottom two decision. Recommend a future rule.

- **Task3: Analyze Pro Dancers and Traits Effects**

  Use the full dataset to model how professional partners and celebrity traits affect outcomes. Compare effects on judges' scores and on estimated *fan vote shares*.

- **Task4: Optimize Competition System**

  Propose a new ranking system that balances fairness and fan engagement. Justify improvements with data driven metrics and a clear adoption rationale.

1.3 Data Cleaning

We identified recurring data-quality issues and applied rule-based cleaning, summarized in Table 1.

- **Missing values.** Judge-score entries equal to zero were set to `NaN`, and a **zero_placeholder** flag was retained.

- **Result parsing.** We parsed the results field into standardized outcome and timing indicators to support consistent modeling across weeks and seasons.

- **Name standardization.** For `ballroom_partner`, we preserved the raw string and extracted a standardized partner name and only one example is provided in the Table 1, while keeping week-specific notes as auxiliary metadata.
- **Reshaping and checks.** Weekly judge scores were converted to a long-format table, then we verified key uniqueness, removed duplicates, and ran basic consistency checks.

Table 1: Representative corrections in data cleaning.

Field	Original	Cleaned
`week*_judge*_score`	0	NaN
`exit_week_inferred`	1	2
`ballroom_partner`	raw string	standardized name

1.4 Our Work

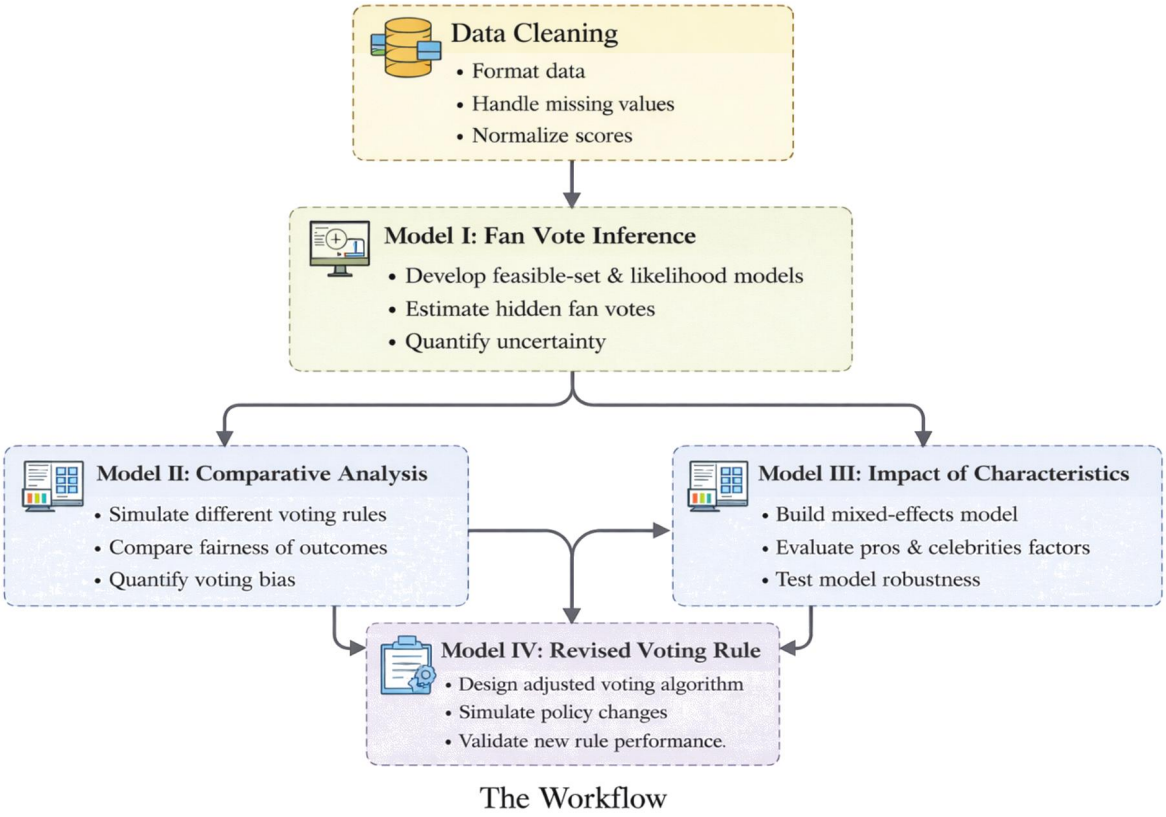


Figure 1: Workflow of our modeling framework.

2 Assumptions

To simplify our modeling framework while retaining the essential mechanisms of the competition, we make the following assumptions. Each assumption will be revisited where it is used.

- **Assumption1** The competition format and the vote–judge aggregation rule are stable within each season, and eliminations follow the stated mechanism with ties resolved consistently.
- **Assumption2** After cleaning, the dataset is reliable and internally consistent; judge-score zeros are post-elimination placeholders treated as missing, and missing judge entries reflect format or schedule differences rather than true zeros.
- **Assumption3** In week t, fan support is modeled by nonnegative vote shares $\mathbf{p}_t = \{p_{i,t}\}_{i=1}^{n_t}$ satisfying $\sum_{i=1}^{n_t} p_{i,t} = 1$. Since the weekly total number of fan votes is unobserved, the model identifies only relative support.
- **Assumption4** Judge evaluations provide an informative signal that is comparable within the same week after normalization; cross-week comparisons rely primarily on within-week ranks or percentiles due to possible score-scale shifts.
- **Assumption5** The inverse problem may admit multiple fan-vote allocations consistent with observed eliminations; therefore, we report uncertainty using credible intervals or entropy rather than claiming a unique true vote total.

3 Notations

Table 2: Key notations used in this paper

Symbol	Definition
$RMSE$	Root mean square error.
R^2_{marg}	Marginal R^2
R^2_{cond}	Conditional R^2
$\tilde{H}$	Mean normalized entropy of weekly fan-share distribution.
$\bar{w}$	Mean feasible-interval width.
δ_{smooth}	Week-to-week smoothness bound
δ_{start}	Initialization bound
CR	Smoothness feasibility conflict rate.
EC_B	Model B elimination consistency rate.
FR_A	Model A feasibility rate.
VR_A	Model A representative point validity rate.
F^P	Fan-share input inferred under Percentage constraints.
F^R	Fan-share input inferred under Rank constraints.
N_{conf}	Number of conflict weeks.

4 Fan Vote Inference via Feasible-Set and Likelihood Models

4.1 Model Overview and Construction

4.1.1 Workflow of Our Models

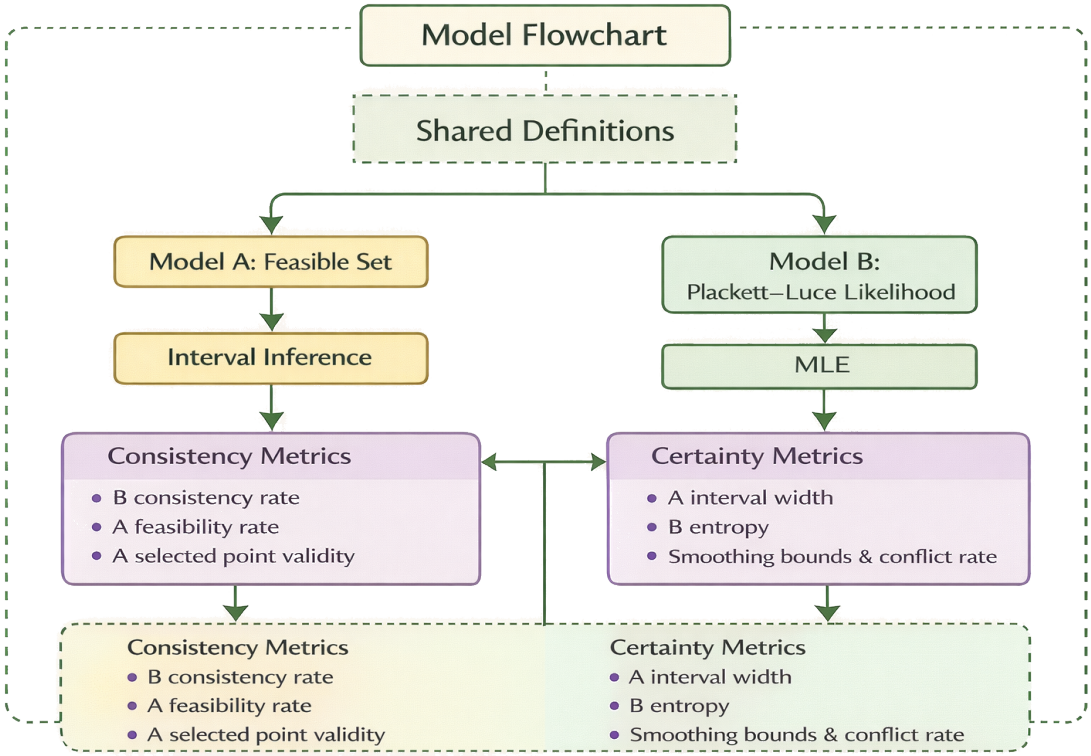


Figure 2: An Overview of Models

4.1.2 Models Building

Because weekly fan votes are not publicly observed, we infer each contestant's weekly fan-vote share from the available weekly judges' scores and elimination outcomes. We introduce two complementary inference approaches and refer to them as Model A and Model B throughout the paper. Both models rely on the same judge-derived signal and the same combined-score construction implied by the official aggregation rule. Model A is a constraint-based interval model that defines a feasible set of fan-share vectors and produces interval estimates using linear constraints [7, 8]. Model B is a likelihood-based point-estimation model that applies a Plackett–Luce likelihood to obtain fan-share point estimates [3, 4, 5].

Shared definitions for Models A and B.

$$j_{i,s,t} = \frac{\exp(\tilde{J}_{i,s,t}/T)}{\sum_{k \in A_{s,t}} \exp(\tilde{J}_{k,s,t}/T)}, \tag{1a}$$

$$C_{i,s,t} = w\, j_{i,s,t} + (1 - w)\, p_{i,s,t}. \tag{1b}$$

Model A: feasible set for interval inference.

$$\mathcal{F}_{s,t} = \left\{ p_{\cdot,s,t} \;\middle|\; p_{i,s,t} \geq 0,\ \sum_{i \in A_{s,t}} p_{i,s,t} = 1,\ C_{e,s,t} \leq C_{k,s,t}\ \forall e \in E_{s,t}, \forall k \in A_{s,t} \setminus E_{s,t} \right\}. \tag{2}$$

Model B: Plackett–Luce likelihood for point inference.

$$\mathbb{P}(e_1, \ldots, e_m) = \prod_{r=1}^{m} \frac{\exp(-\kappa C_{e_r,s,t})}{\sum_{k \in R_r} \exp(-\kappa C_{k,s,t})}. \tag{3}$$

Table 3: Notation used in (1a)–(3).

Symbol	Definition
$s,\ t,\ i$	Season, week, and contestant indices.
$A_{s,t},\ E_{s,t}$	Active set and eliminated set.
$p_{i,s,t}$	Fan-vote share and its estimate.
$\tilde{J}_{i,s,t},\ j_{i,s,t}$	Standardized judge score and softmax judge signal.
$C_{i,s,t}$	Combined score for elimination risk.
$T,\ w,\ \kappa$	Temperature, mixture weight, and likelihood sharpness.
R_r	Remaining set before the r-th elimination.
$p_{\cdot,s,t}$	Vector of fan shares over $i \in A_{s,t}$.
$m,\ e_r$	Number of eliminations and the r-th eliminated contestant.

The feasible-set representative point follows a maximum-entropy criterion grounded in Shannon entropy [9].

4.2 Diagnostic Plot for Judge Signal and Fan Share

We run the inference pipeline once on the cleaned weekly dataset to produce a diagnostic view of how judge information relates to the fan component implied by eliminations. For each week we construct a within week judge signal from judge totals using two definitions, percent based and rank based, standardize within the week, and convert to shares via a softmax transform. We then

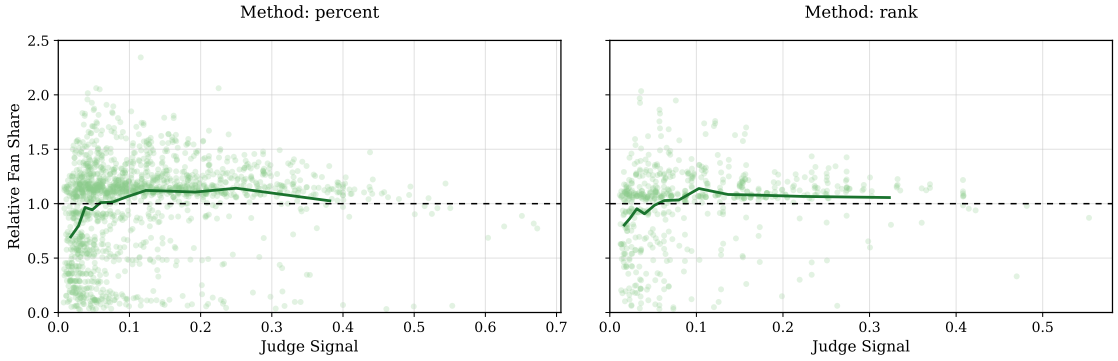


Figure 3: Relationship between judge signal and relative fan share

fit a preliminary Model B to infer fan shares for active contestants and form relative fan share by multiplying by the number of active contestants. Figure 3 plots judge signal against relative fan share with a binned mean curve.

The binned trend rises slightly at low to mid signal, then levels off and mildly declines at higher signal, while dispersion remains large. This suggests judge scores alone do not deterministically represent fan support, so we use a soft combination of judge signal and a latent fan component and rely on elimination constraints for inference with uncertainty reporting. The similarity across the two signal definitions indicates the pattern is robust.

4.3 Validation and Uncertainty Quantification

4.3.1 Consistency metric

We check whether fan share estimates inferred under a specified aggregation rule can reproduce the observed weekly eliminations. A week is counted as consistent for a given rule if the contestants in $E_{s,t}$ appear among the lowest combined scores computed using the fan shares inferred under that same rule.

Table 4 reports three internal diagnostics that separate elimination consistency, constraint feasibility, and sensitivity to the representative point choice. These diagnostics quantify whether the inferred shares satisfy the elimination logic and whether a selected representative solution remains consistent within the feasible set. They do not measure predictive accuracy because the weekly eliminations are part of the information used to infer fan shares. Specifically, EC_B is the share of weeks where the Model B point estimate reproduces the observed eliminations, FR_A is the share of weeks where the Model A constraints admit at least one feasible solution, and VR_A is the share of feasible weeks where the selected representative solution also reproduces eliminations.

Agreement is high across all metrics. Model B matches eliminations in **258 of 264** weeks. Model A is feasible in **255 of 264** weeks, and among feasible weeks the representative solution remains consistent in **252 of 255** weeks. The small gap between feasibility and selected point validity suggests that ambiguity within the feasible set rarely changes the implied elimination order. Most infeasible weeks coincide with withdrawals or weeks without eliminations, so excluding these special cases yields feasibility in all remaining weeks.

Table 4: Feasibility and internal consistency diagnostics for elimination constraints.

Metric	Rate	Success Total
EC_B	97.7%	258 of 264
VR_A	98.8%	252 of 255
FR_A	96.6%	255 of 264

4.3.2 Certainty metric

We quantify identification uncertainty using interval width from Model A and dispersion using normalized entropy from Model B.

$$\bar{w} = \text{mean}\big(U_{i,s,t} - L_{i,s,t}\big) \tag{4}$$

$$\bar{\tilde{H}} = \text{mean}\left(\frac{-\sum_{i=1}^{n_{s,t}} p_{i,s,t} \log p_{i,s,t}}{\log n_{s,t}}\right) \tag{5}$$

Symbols. L_{ist} and U_{ist} are feasible lower and upper bounds of Model A fan share for contestant i in season s week t. p_{ist} is the Model B point estimate and n_{st} is the number of active contestants. The mean operator averages $U_{ist} - L_{ist}$ over all feasible contestant week pairs and averages normalized entropy over all eligible weeks.

Across feasible contestant week pairs, interval width has P10 **0.1226**, P50 **0.1226**, and P90 **0.1557**, while normalized entropy across weeks has P10 **0.8429**, P50 **0.9542**, and P90 **0.9901**.

Table 5: Certainty metrics

Metric	Value	Weeks
$\bar{w}$	0.1267	255
$\tilde{H}$	0.9344	264
δ_{smooth}	0.0619	264
δ_{start}	0.0780	264
CR	0.00046	264

4.3.3 Cross rule robustness check

Because the elimination constraints depend on the aggregation rule, fan share inference is rule conditional. To reduce circularity in later rule comparisons, we infer two fan share sets for every week. One set is inferred under Rank constraints and the other set is inferred under Percentage constraints. In Section 5, we cross apply these two inputs when auditing rules and report whether directional conclusions persist under both inference settings.

4.3.4 One week ahead predictive check

To validate predictive performance without using the current week elimination outcome, we conduct a one week ahead check. Eligible weeks are those with a single elimination, complete $t-1$ fan share estimates for all contestants active in week t, and no withdrawals or multi-elimination formats. For each eligible week, we take the inferred fan shares from week $t-1$, restrict to contestants active in week t, renormalize shares to sum to one, and combine with the observed judges totals to predict the eliminated contestant under each rule.

We evaluate predictive quality using two metrics. Hit rate is the fraction of weeks where the predicted elimination matches the true elimination. Average eliminated predicted rank is the average rank position of the true eliminated contestant under the predicted combined score ordering, where rank one denotes the lowest predicted combined score.

We use 213 eligible weeks with available week $t-1$ shares for all active contestants. Under the Percentage rule, the hit rate is **98.59** percent and the average eliminated predicted rank is **1.0141**. Under the Rank rule, the hit rate is **78.87** percent and the average eliminated predicted rank is **1.1737**. This check should be viewed as a limited sanity test rather than a fully out-of-sample forecast because the week $t-1$ shares are inferred under rule-dependent constraints.

Table 6: One week ahead predictive check summary

Rule	Weeks	Hit rate percent	Avg eliminated predicted rank
Percentage	213	98.59	1.0141
Rank	213	78.87	1.1737

5 Voting Mechanism Audit via Counterfactual Analysis

5.1 Week-by-Week Counterfactual Simulator

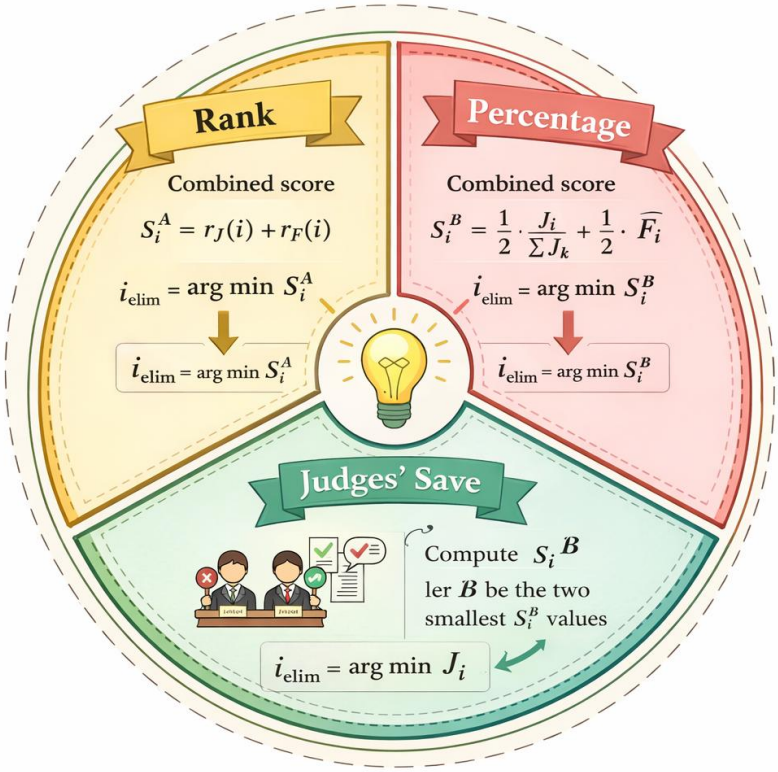


Figure 4: Three elimination rules evaluated in our counterfactual simulator

We build a week-by-week counterfactual simulator to compare three elimination rules under the same information set, namely the Rank rule, the Percentage rule, and the Judges' Save rule. For each week indexed by season s and week t, we keep the active contestant set fixed and use the observed judges' totals $J_{i,s,t}$. To avoid circularity in rule comparison, we prepare two fan-share inputs for each week, denoted F^R and F^P, inferred under Rank constraints and Percentage constraints, respectively. For each input, we recompute which contestant would be eliminated under each of the three rules and summarize agreement and conflict diagnostics. [11]. All rule-comparison statistics in this section use a fixed analysis set of eligible weeks with single elimination and complete judges' totals after cleaning, excluding no-elimination, multi-elimination, and withdrawal weeks, yielding $N = 264$ analysis weeks.

Within each week, we use within-week ranks to standardize scales, where rank one denotes the best standing and ties receive the average rank. Let J_i denote the judges' total and let F_i denote the inferred fan share for contestant i in that week. Let $r_J(i)$ and $r_F(i)$ be the within-week ranks of J_i and F_i, and let S_i denote the rule-specific elimination-risk score. Under the Judges' Save rule, we first identify the bottom two contestants under the Percentage combined score and then eliminate the contestant with the lower judges' total among those two. Figure 4 summarizes the three rules.

5.2 Key Evidence of Fan Amplification

To quantify how strongly a rule aligns with fan preferences, we compute a weekly *fan-bias coefficient* as the Spearman correlation between the combined score and the estimated fan share [6]:

$$\rho_{\text{fan}} = \text{Spearman}(\{S_i\}, \{F_i\}).$$

All metrics in this section are computed under both fan share inputs described in Section 5.1. We report paired values for F^P and F^R, and Table 7 summarizes the cross-inference audit.

Larger values of ρ_{fan} indicate that the rule tracks fan votes more closely, meaning popularity has greater influence on outcomes.

We also track conflict weeks, defined as weeks where Rank and Percentage eliminate different contestants given the same inputs. To reduce circularity in rule comparison, we compute all key metrics under both fan share inputs and summarize the audit in Table 7.

Table 7: Cross inference audit summary

Input	ρ_{fan} Rank	ρ_{fan} Percent	N_{conf}	Percent wins
F^P	0.6998	0.7770	50	50 of 50
F^R	0.6976	0.7827	50	50 of 50

Table 7 shows that the rule comparison conclusions are stable across inference inputs.

Across all $N = 264$ analysis weeks, both fan-share inputs yield consistent conclusions. Paired values (F^P/F^R) are:

- **Higher fan alignment under Percentage:** The average fan-bias coefficient ρ_{fan} equals **0.7770/0.7827** under Percentage, versus **0.6998/0.6976** under Rank. Details are shown in Figure 5.
- **Meaningful outcome disagreement:** The Rank and Percentage rules produce different eliminations in **50 out of 264** analysis weeks under both inputs, corresponding to an **18.94%** conflict rate.
- **Directional evidence in conflicts:** In every conflict week, the Percentage rule retains the contestant with the higher fan share, yielding **50 of 50** under both inputs. This empirical pattern arises because Rank discards magnitude information, so disagreements are typically driven by cases where Rank eliminates a higher fan share contestant due to an extreme judge rank. We further note that this pattern depends on the score and vote distributions in our dataset and may not hold in other datasets.

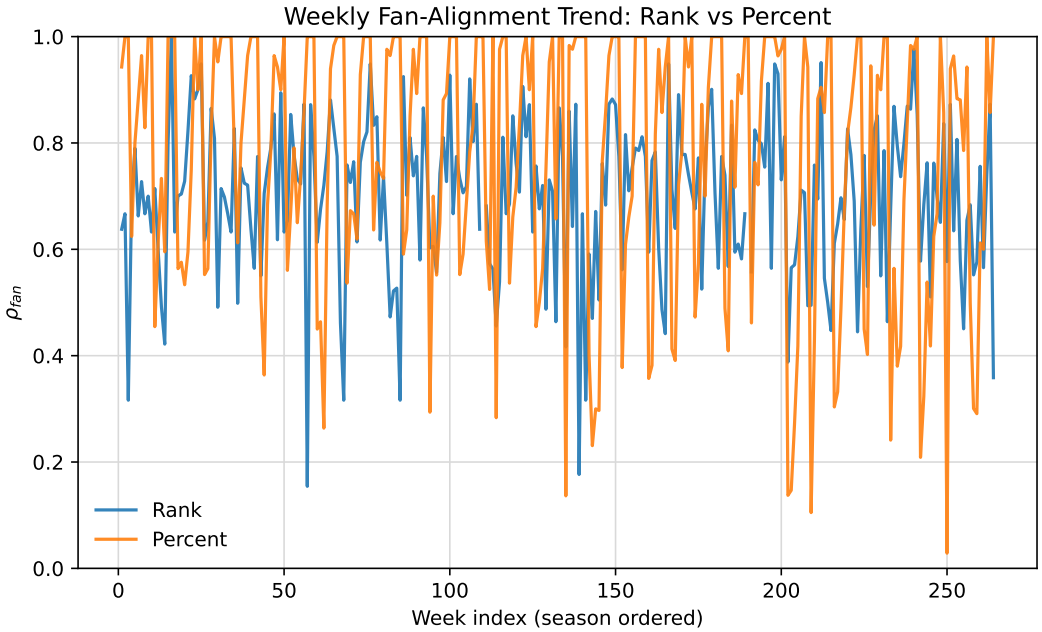
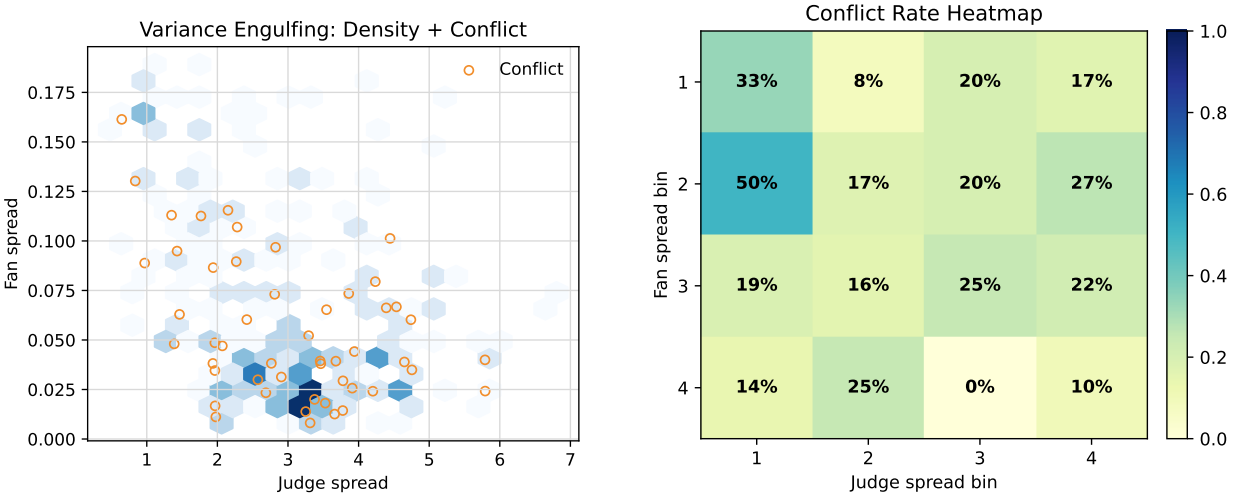


Figure 5: Weekly Fan-Alignment Trend: Rank vs Percent

5.3 Why Percentage Flips Eliminations: "Variance Engulfing"

We observe a practical pattern in conflict weeks. Judges' totals are often tightly clustered, meaning the spread in judges' scores is small, while fan shares are much more uneven, meaning the spread in fan support is large. Because the Percentage method adds the two shares directly, even a moderate gap in fan votes can outweigh a substantial deficit in judges' scores and reverse who gets eliminated. We refer to this mechanism as *variance engulfing*.



(a) Variance Engulfing: density with conflict

(b) Conflict rate heatmap by spread bins

Figure 6: Variance Engulfing evidence from weekly spread patterns

Representative example from Season 1, Week 2 illustrates the mechanism. Evander Holy-

field had the worst judges' result that week, yet his estimated fan share was relatively strong at approximately **9.4%**. Trista Sutter placed better with the judges, but her fan share was low at approximately **2.0%**. Under the Rank rule, Evander is eliminated because the last-place judges' rank becomes decisive after rank normalization. Under the Percentage rule, Trista is eliminated because the fan-share gap of approximately **7.4** percentage points dominates the combined share.

5.4 Controversial Figures: What Changes Under Alternative Rules?

To avoid purely anecdotal discussion, we ran the same week-by-week counterfactual protocol for four frequently debated contestants. The results can be summarized succinctly:

- **Billy Ray Cyrus (S4):** under Rank and under Judges' Save, he would exit earlier, suggesting survival benefited from Percentage-style aggregation.
- **Jerry Rice (S2) and Bristol Palin (S11):** outcomes remain broadly stable under the tested rules, consistent with non-pathological trajectories.
- **Bobby Bones (S27):** remains safe under all tested standard rules because he rarely enters Bottom2; therefore Judges' Save is not activated and cannot correct the outcome.

We implement Judges' Save deterministically as eliminating the worse-judged contestant within Bottom2. The real show uses judge discretion, so these counterfactuals represent one plausible, conservative implementation rather than the only outcome.

5.5 Policy Recommendation

Based on the diagnostics above, we recommend a two-step reform:

1. **Return to Rank aggregation .** This reduces structural fan amplification and lowers conflict-driven flips.
2. **Strengthen judge intervention beyond Bottom2 when needed.** Since Bottom2-based Judges' Save may never activate for extreme fan-support cases, expanding the review pool increases the chance that judge review can meaningfully operate.

6 Impact of Pros and Celebrity Characteristics: Mixed-Effects Model

6.1 Data unit & outcomes

To study how pros and celebrity traits affect performance, we form a week level pair panel indexed by season s, week t, celebrity c, and professional dancer p. The cleaned dataset provides the active pairs and their weekly judges total. To incorporate fan support, we merge the fan vote share estimates from Model I into this panel using season, week, and the celebrity identifier.

We model two outcomes. For judges, we use the judges total standardized within each season, denoted judge z. For fans, we use the Model B point estimate $p_{i,s,t}$ as an inferred fan share signal. Because $p_{i,s,t}$ is scale free and the number of active contestants varies by week, we construct a fan support index

$$\text{fan rel}_{i,s,t} = \log p_{i,s,t} + \log n_{s,t} \tag{6}$$

where $n_{s,t}$ is the number of active contestants in week s, t. This index adjusts for week size, so it approximates relative popularity on a comparable scale; it should be interpreted as relative support rather than an absolute vote count. We then standardize fan rel to unit variance for coefficient comparison.

6.2 Model specification

We use a compact mixed-effects model [12]. Let y_{it} denote the outcome for week t and pair i in season s. The pair is a celebrity and a professional dancer. The judges outcome is

$$y^J = \beta_0^J + \mathbf{x}^\top \boldsymbol{\beta}^J + \alpha_s + u_p + u_c + \varepsilon, \tag{7}$$

and the fans outcome is

$$y^F = \beta_0^F + \mathbf{x}^\top \boldsymbol{\beta}^F + \alpha_s + u_p + u_c + \varepsilon. \tag{8}$$

The variables and parameters used in Equations 7 and 8 are summarized in Table 8.

The fan shares $p_{i,s,t}$ are inferred quantities rather than direct observations. In the mixed effects fit, we use point estimates as inputs and do not model measurement error in $p_{i,s,t}$. This choice can understate uncertainty in fixed effect estimates and produces confidence intervals that may be optimistic.

Table 8: Key notations for the mixed effects model

Symbol	Definition
s, t, i, c, p	Season, week, pair, celebrity, and professional dancer indices.
y^J, y^F	Judges outcome and fans outcome for a given season s, week t, and pair i.
$\mathbf{x}$	Covariate vector for the same pair week observation.
β_0^J, β_0^F	Fixed intercepts for judges and fans models.
$\boldsymbol{\beta}^J, \boldsymbol{\beta}^F$	Fixed effect coefficient vectors for judges and fans models.
α_s	Season fixed effect.
u_p, u_c	Pro and celebrity random intercepts with mean zero.
ε	Week level disturbance term.

6.3 Mixed effects results and interpretation

We first summarize overall model fit to set the scale of effects. Table 9 reports RMSE and R^2 for both outcomes. For each outcome, R^2_{cond} is larger than R^2_{marg}, following the mixed-model interpretation of marginal versus conditional explained variance [13]. This means persistent differences across pros and celebrities explain a lot beyond observed covariates.

Table 9: Model fit summary for the mixed-effects specifications.

Outcome	RMSE	R^2_{marg}	R^2_{cond}
Judges	0.6075	0.2376	0.6035
Fans	0.5740	0.1324	0.3202

6.3.1 Professional dancer effects

The random effects u_p quantify pro-specific net contributions after controlling for celebrity traits and season and week context. These estimates shrink toward zero when data are sparse, so rankings are more reliable for pros who appear in more weeks. In our results, pro effects are more pronounced for judges than for fans, which suggests pros influence judges' scores more strongly than they influence fan voting.

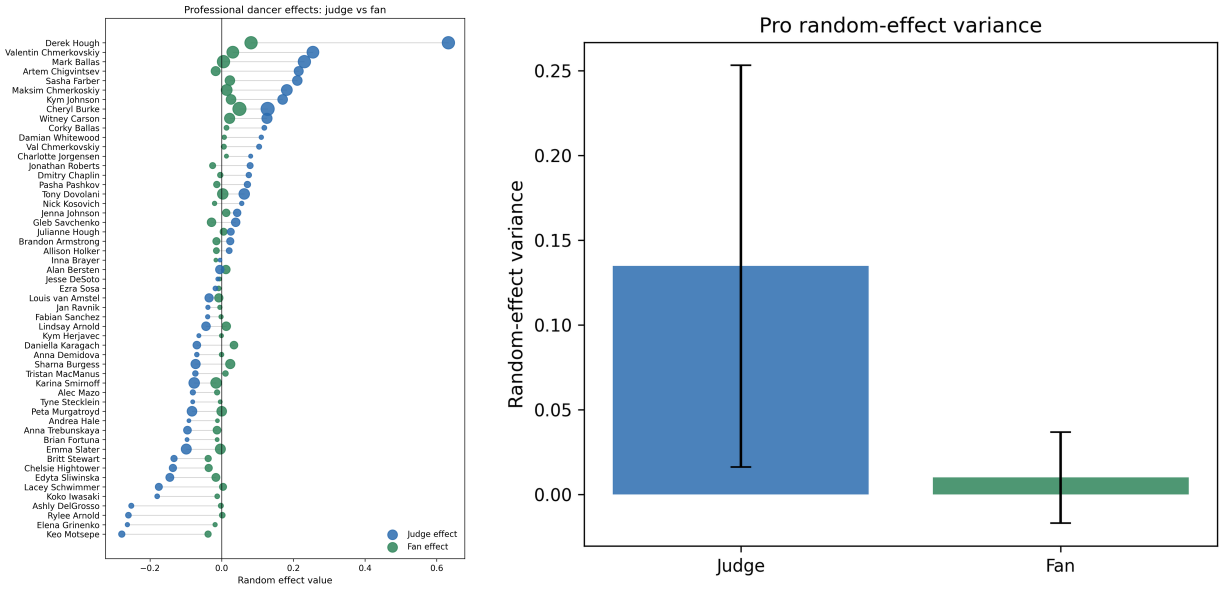


Figure 7: Professional dancer effects are stronger for judges than for fans

The Figure 7 directly quantifies how much different pros matter, and it shows that pro effects are stronger for judges outcomes than for fans outcomes.

6.3.2 Celebrity characteristics

To assess whether measurable celebrity traits affect both judges outcomes and fan outcomes, we use the same covariate set in the two outcome equations and compare the direction and magnitude across outcomes.

For age related effects, we standardize age to make coefficients comparable across variables. We find a negative age coefficient for both outcomes, suggesting that after controlling for season effects, older contestants tend to receive lower judges scores and attract weaker fan support.

For industry background, we encode industry as indicator variables relative to a baseline category. We find that some categories are negative for both judges and fans, such as Model, while other categories are weaker or differ in direction across outcomes, indicating that industry effects can vary by evaluation channel.

For competition context, we include the week covariate and the number of judges to capture systematic changes over the season. We find a negative week effect in both models, consistent

with later weeks being harder on average and with fan support becoming more concentrated as the season progresses. In contrast, the num judges effect is smaller and less stable.

Table 10 summarizes the key fixed effects with 95% confidence intervals, and the full table is provided in Appendix.

Table 10: Key fixed effects

Variable	Judges: $\hat{\beta}$	Fans: $\hat{\beta}$
Age	−0.475 [-0.547, -0.402]	−0.254 [-0.318, -0.190]
Week	−0.081 [-0.093, -0.068]	−0.122 [-0.139, -0.105]
Num of judges	0.051 [-0.054, 0.157]	0.096 [-0.041, 0.234]
Model	−0.543 [-0.927, -0.159]	−0.703 [-1.059, -0.346]
TV personality	−0.438 [-0.648, -0.229]	−0.090 [-0.275, 0.095]
Athlete	−0.264 [-0.466, -0.061]	0.008 [-0.170, 0.186]
Social media	0.033 [-0.462, 0.527]	−0.246 [-0.674, 0.181]
Other	−0.311 [-0.591, -0.031]	0.033 [-0.218, 0.284]

Comparing coefficients across outcomes shows that some traits move judges and fans in the same direction, while others differ in magnitude or direction, so the two outcomes do not respond in the same way to every characteristic.

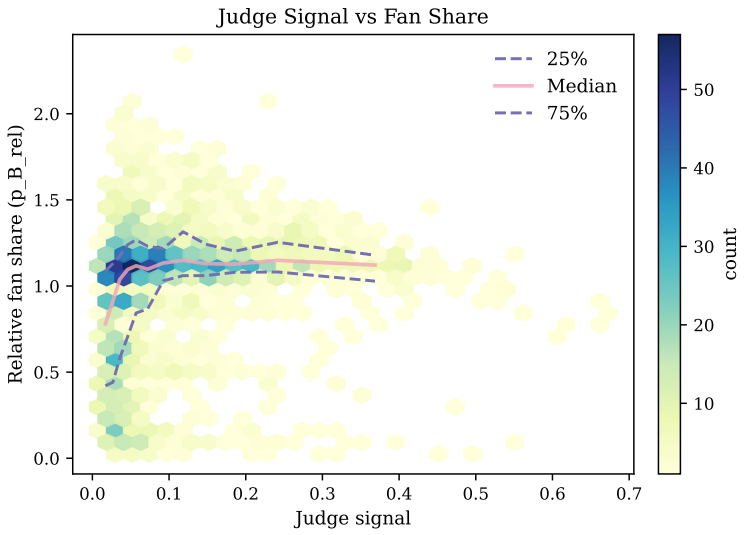
6.3.3 Do impacts differ between judges and fans?

We compare judges outcomes and fan outcomes to assess whether the same characteristics act in the same way. Figure 8b suggests a positive trend between judge signal and relative fan share, which we quantify in Table 8a.

Metric	Est	CI lo	CI hi
Pearson r	**0.167**	0.131	0.202
Spearman ρ	**0.262**	0.223	0.297

Note: Bootstrap uses 2000 resamples with seed 0 and n = 2249.

(a) Correlation summary



(b) Hexbin density plot

Figure 8: Judge signal versus relative fan share

Table 10 supports this pattern. Some traits align across outcomes, such as age being negative for both, while others differ in magnitude or direction. Pro random effects are also more pronounced for judges than for fans, reinforcing that the two mechanisms respond differently to certain factors.

6.4 Robustness checks

We run two robustness variants to test whether our conclusions depend on outcome construction. First, we re-estimate the fan outcome using an alternative vote-share estimate. Second, we standardize judges' scores within each week rather than within each season. Figure 9 reports the absolute coefficient differences between the baseline and robustness runs for age and key industry groups. The changes are uniformly small, indicating that the qualitative conclusions about age and key industry effects remain stable. This strategy is consistent with standard robustness and validation practice in statistical learning [14, 10].

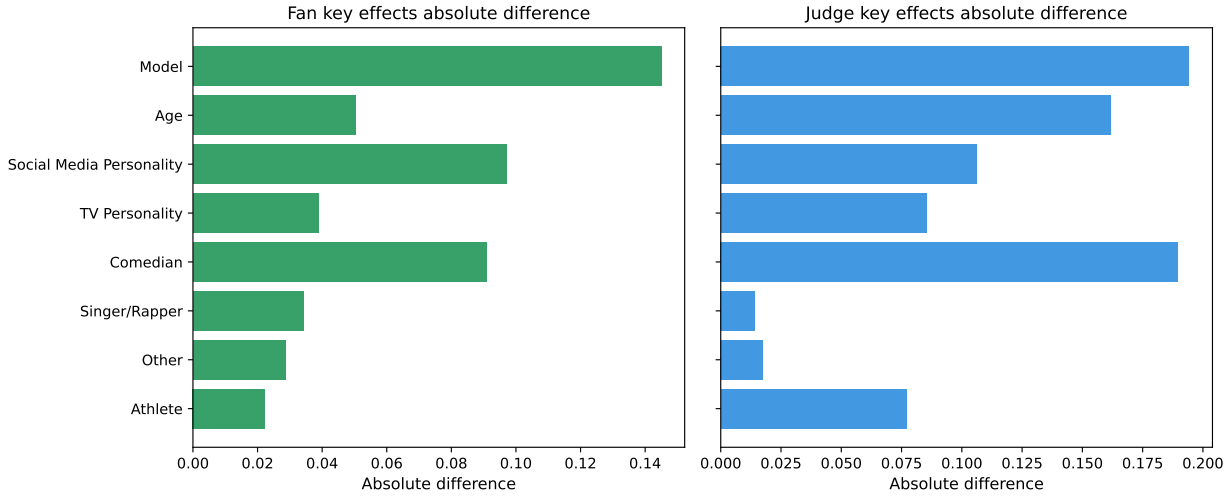


Figure 9: Key-coefficient absolute differences under robustness checks

7 Proposed Weekly Voting Rule: MARS-Soft+Disagreement

7.1 Design Rationale and Mechanism Overview

We seek a weekly voting rule that preserves the transparency and stability of rank aggregation while addressing a recurring concern in contested weeks: judges may strongly favor a contestant whose fan support is comparatively weak. Our goal is to introduce only a minimal, targeted correction that mitigates judge–fan mismatches without over-amplifying popularity. The rule should be simple to implement, scale-aware across weeks with different contestant counts, and retain meaningful fan influence in determining eliminations. [11].

Figure 10 summarizes the overall workflow of the proposed mechanism. Each week, we first collect the required inputs from the same information set, including judges' totals and inferred fan shares for the active contestants. We then generate a baseline elimination outcome under a transparent rank-aggregation baseline. Next, we assess the alignment and disagreement between judges and fans to identify weeks and contestants where a targeted adjustment is warranted. We apply a small voting adjustment that is designed to be selective and limited in magnitude. Finally, we recompute the weekly elimination after the adjustment and record the resulting outcome.

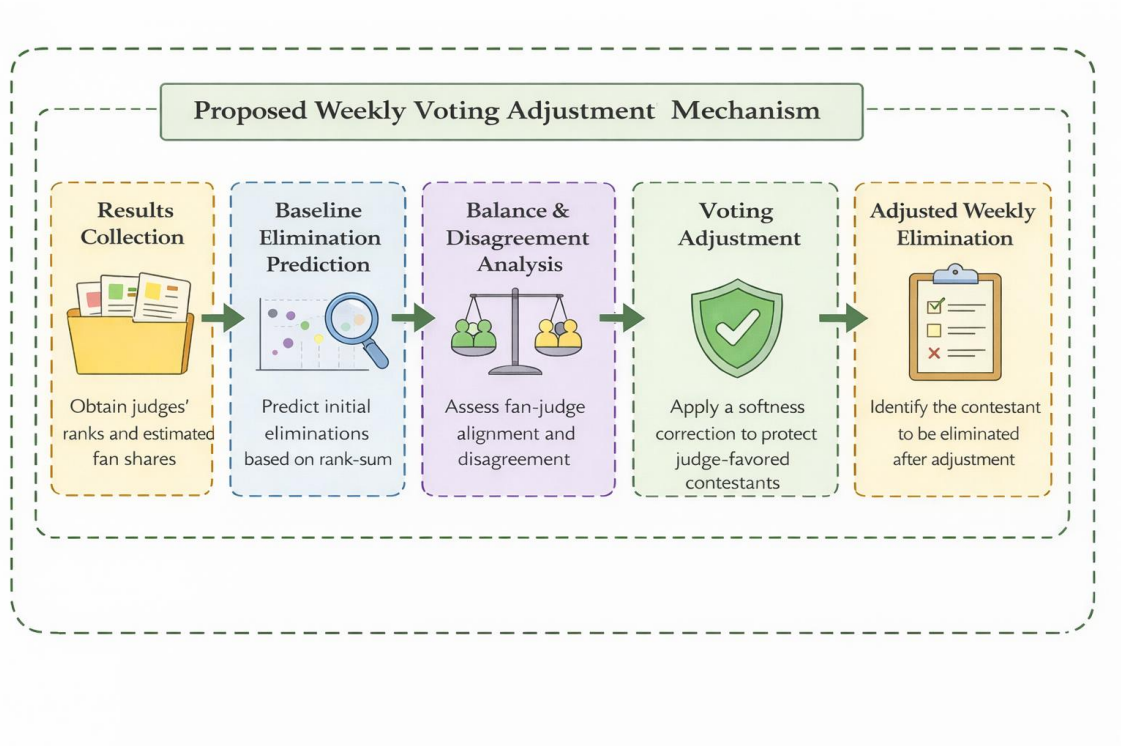


Figure 10: Overview of the proposed voting adjustment mechanism

7.2 Rule definition

Within each week, let $R_J(i)$ and $R_F(i)$ be the ranks of judges' totals and fan shares, with rank 1 = best (ties averaged). Ranks denote performance ordering and are used to construct the MARS risk score. The base score is

$$S_i^0 = R_J(i) + R_F(i), \tag{9}$$

where larger values indicate worse standing. We then add a gated, soft protection term that activates only for judge-elite contestants who are ranked worse by fans:

$$q_i = \frac{R_J(i) - 1}{n - 1}, \qquad g_i = \left(1 - \frac{q_i}{\alpha}\right)_+^p, \qquad d_i = \frac{\max(0,\ R_F(i) - R_J(i))}{n - 1}, \tag{10}$$

$$S_i = S_i^0 - \lambda_t\, g_i\, d_i, \qquad \lambda_t = \beta n. \tag{11}$$

Define

$$x_+ = \max\{x, 0\} \tag{12}$$

When a contestant is not in the top α fraction by judges or when judges and fans agree ($R_F(i) \le R_J(i)$), the bonus vanishes and the rule reduces to the rank-sum baseline. The weekly elimination is the contestant with the largest S_i, and ties are broken by higher elimination risk under judges first, then fans, then lower raw judges total, then lower fan share.

7.3 Evaluation protocol and metrics

Evaluation protocol

We use the same week-by-week counterfactual design as Section 5.1. For each historical week, we keep the observed judges' totals and estimated fan shares, then compare the MARS elimination with the baseline Percentage rule.

Metrics

- **RescueRate**: fraction of baseline-eliminated contestants who are in judges' top-3 but are not eliminated by MARS.
- **Balance gap** $\Delta\rho$: average $|\rho_J - \rho_F|$, where ρ_J and ρ_F are Spearman correlations between S_i and the judge/fan ranks, respectively.
- **FII**: Fan Impact Index, the rate at which small fan-share perturbations change the elimination.
- **ChangeRate**: fraction of weeks where MARS changes the baseline elimination set.

Parameter choice

We split seasons into training data from S1 to S17 and testing data from S18 to S34. We then perform a grid search over α in $\{0.20, 0.25, 0.30\}$, β in $\{0.20, 0.30, 0.40\}$, and p in $\{1, 2\}$, following standard hyperparameter selection practice [14]. The best setting is

$$\alpha = 0.20, \qquad \beta = 0.20, \qquad p = 2.$$

7.4 Results and discussion

Results

Table 11: Summary performance of MARS-Soft+Disagreement

Split	Weeks	RescueRate	$\Delta\rho$	ChangeRate
Train	134	0.588	0.186	0.299
Test	130	0.789	0.210	0.385

Note. Test split FII = 0.161.

Overall, the rule rescues judge-elite contestants when disagreement is large, keeps judge–fan balance stable with a balance gap around **0.19–0.21**, and does not cause excessive changes with ChangeRate around **0.30–0.39**.

Recommendation and limitations

We recommend *MARS-Soft+Disagreement* as a simple weekly rule that is scale-fair, protects top technical performers only when such protection is warranted, and still preserves meaningful fan influence. The rule is straightforward to compute and practical to implement in real settings.

Our analysis also has important limitations. Week-by-week counterfactuals cannot reconstruct full-season trajectories unless future judges' scores are imputed, so our results should be interpreted as local, week-level comparisons rather than end-to-end season replays. In addition, the Fan Influence Index depends on the assumed uncertainty model for fan shares, so its magnitude is best used for comparative evaluation across rules instead of as an absolute measure.

8 Model Sensitivity Analysis

8.1 Fan-Vote Inference

Initial values and sensitivity ranges

We vary judge-weight, softmax temperature, and cross-week smoothing bounds around calibrated values, following standard local perturbation logic in sensitivity analysis [14, 10].

Table 12: Initial values and sensitivity ranges for fan-vote inference.

Parameter	Baseline	Lower	Upper
W_{judge}	0.30	0.30	0.70
TEMP_K	5.0	2.0	20.0
δ_{smooth}	0.0619	0.0371	0.0866
δ_{start}	0.0780	0.0468	0.1093

Results and observations

When W_{judge} increases, consistency decreases. Specifically, as W_{judge} rises from **0.3** to **0.7**, consistency drops from **0.947** to **0.739** and then to **0.496**, indicating that the estimate is most sensitive to the judge weight. A higher `TEMP_K` increases consistency; for instance, consistency is **0.826** at `TEMP_K` = 2 and **0.996** at `TEMP_K` = 20. The δ-grid exhibits a smooth trade-off, and $\delta_{\text{smooth}} \approx$ **0.0619** lies in a flat, feasible region. Overall, temperature and smoothing parameters mainly shift uncertainty bounds without breaking feasibility.

8.2 Pros & Traits Impact

Initial values and sensitivity ranges

We change outcome definitions by comparing week-standardized with season-standardized judges' scores and by comparing p_B-based with p_A-based fan outcomes, and we use season-blocked validation. This tests whether conclusions are stable under plausible specification choices [10].

Results and observations

Table 13: Robustness summary for pros and traits

Item	Judges	Fans
RMSE — In-Sample	0.607	0.574
RMSE — Season CV	0.923	0.716
EC — In-Sample	0.364	0.580
EC — Season CV	0.227	0.258
$\mathbf{R}^2_{\text{base}}$	0.238 / 0.604	0.132 / 0.320
$\mathbf{R}^2_{\text{rob}}$	0.537 / 0.719	0.068 / 0.310

Note. $\mathbf{R}^2$ *is reported as* R^2_{marg}/R^2_{cond}. *In-sample metrics are computed on the data used for fitting, while Season CV uses season-blocked cross-validation that holds out entire seasons for evaluation.*

Table 13 shows that conditional R^2 is much larger than marginal R^2 under both baseline and robust outcomes, implying that random effects for pros and celebrities consistently explain substantial variance. Season-blocked validation increases RMSE to **0.923** for judges and **0.716** for fans and reduces elimination consistency, reflecting degraded generalization when entire seasons are held out.

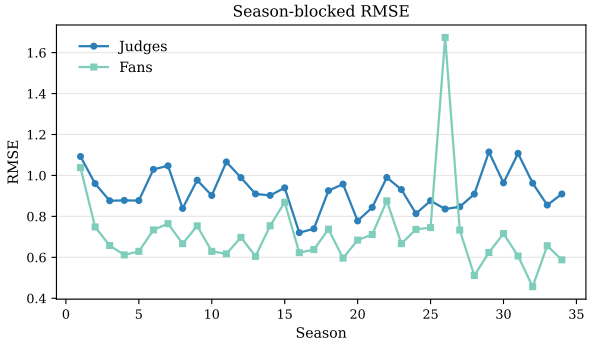


Figure 11: Season-blocked RMSE by season

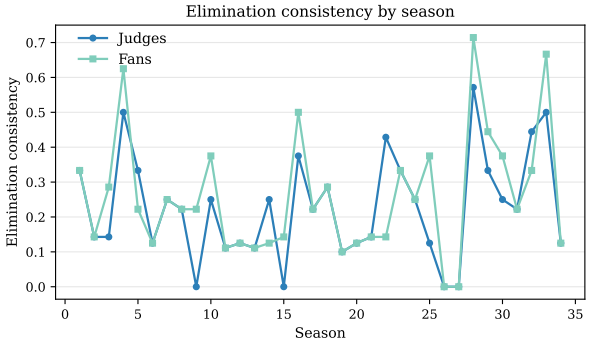


Figure 12: Elimination consistency by season

To further support these aggregate results, Figure 11 plots season-wise RMSE and Figure 12 plots season-wise elimination consistency under season-blocked validation. Both figures exhibit clear cross-season variability, and several seasons show simultaneously higher RMSE and lower consistency, matching the direction of change summarized in Table 13.

Overall, the season-level breakdown corroborates the robustness summary: age remains negative, and the week coefficient is most sensitive to normalization.

9 Overall Model Evaluation

9.1 Strengths

1. **Reliable week-level reconstruction.** The inference framework consistently recovers elimination patterns and yields stable week-level fan-support estimates, enabling transparent analysis when votes are unobserved.

2. **Clear mechanism diagnosis for rule behavior.** The counterfactual audit isolates how aggregation rules differ and provides consistent directional evidence on when and why fan influence is amplified.

3. **Practical policy evaluation and interpretability.** The proposed rule is simple, predictable under disagreement, and enables fairness comparisons without full-season simulation.

9.2 Weaknesses

1. **Limited cross-season transferability.** Performance degrades under season-blocked settings, suggesting temporal instability across eras and the need for recalibration when forecasting across seasons.

2. **Sensitivity to modeling choices for uncertainty.** Metrics that rely on the assumed perturbation or uncertainty model should be viewed as relative indicators for comparing rules, not as absolute measurements.

3. **Dependence on the assumed aggregation rule.** Fan-share inference is rule-conditional; we mitigate this via cross-inference under Rank and Percentage, but some structural sensitivity remains.

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Memorandum

To: DWTS producer

From: MCM Team 2627351

Subject: Recommendations and proposed solutions based on judges and fan votes

Date: February 2, 2026

Dear Madam or Sir,

This memorandum provides evidence-based recommendations for improving the fairness and transparency of weekly eliminations while preserving meaningful audience influence. Because fan votes are not directly observed, we combine the published judges' totals with statistically inferred fan-support signals to audit how different aggregation rules behave under the same information set. We focus on practical questions that matter to viewers and producers, including when outcomes become sensitive to popularity, when rules disagree on who should leave, and how often those disagreements would change eliminations.

Our analysis is designed to support decision-making rather than to replace creative or production judgment. We summarize the key patterns that consistently appear across weeks and seasons, translate them into measurable weekly indicators, and propose a minimal-change adjustment that can be piloted without redesigning the entire format. The recommendations below are organized to offer a preferred path and a conservative fallback, along with a short implementation checklist and risk mitigations to support clear communication with the public.

Key findings.

- **Fan-vote inference is usable for week-level decisions.** Feasible reconstructions exist for almost all weeks, and the inferred shares are consistent with observed eliminations in the vast majority of cases. Uncertainty remains material, which supports transparent reporting rather than hidden fan totals.
- **Percentage aggregation tends to amplify fan influence.** Conflicts with rank-based outcomes occur regularly, and when judge scores are close the percentage method typically favors higher-fan-share contestants.
- **The proposed rule improves judge-elite protection without removing fan impact.** On held-out seasons it rescues more judge-top contestants while keeping disruption moderate and fan influence meaningful.
- **Pro/celebrity effects are persistent.** Mixed-effects results indicate stable partnership and contestant heterogeneity beyond observable traits.

Primary recommendation.

Adopt **MARS-Soft+Disagreement** as the weekly elimination rule. It preserves the rank-based backbone but adds a small, gated protection only when judges strongly favor a contestant who is under-supported by fans.

If you only do one thing. Run a **one-season pilot** of MARS-Soft+Disagreement and publish a short weekly transparency report (fan-alignment, conflict flags, rescue events).

Minimal-change fallback (if rule changes must be conservative).

- Return to **Rank aggregation** as the default (reduces structural fan amplification).
- Expand judge review from **Bottom2** to **Bottom3** so judge intervention can activate when fan shocks are extreme.

Implementation checklist (MARS).

1. Compute weekly ranks of judges and fan shares (rank 1 = best).
2. Apply the MARS score with calibrated parameters in Section 7.
3. Eliminate the largest S_i (risk score), with deterministic tie-breaks: worse judge rank, worse fan rank, lower judge total, lower fan share.
4. Track four weekly KPIs: RescueRate, $\Delta\rho$, ChangeRate, and FII.

Risks and mitigations.

- **Perceived reduction of fan power.** Mitigate by publishing fan-impact metrics to show the fan vote remains influential.
- **Edge cases (multi-elimination weeks).** Use the same rule but eliminate the top-k largest S_i; disclose any special weeks.
- **Public skepticism.** Pair the pilot with a short explainer and a simple dashboard: "what changed this week and why."

We hope that our model can assist decision-makers in evaluating voting mechanisms, improving fairness, and designing more transparent and robust weekly rules. While no rule can eliminate uncertainty, careful analysis and principled design can reduce bias and enhance audience trust. We wish you success in balancing fairness, entertainment, and public engagement.

Sincerely,

Team 2627351

Appendices

Full Fixed-Effects Results for the Mixed-Effects Model

Table 14: Full fixed effects comparison, standardized outcomes. p_J and p_F are two-sided Wald-test p-values for fixed effects in the judges and fans models.

Variable	Judges: β, 95% CI	Fans: β, 95% CI	p_J	p_F
Age	−0.475 [−0.547, −0.402]	−0.254 [−0.318, −0.190]	< 0.001	< 0.001
Week	−0.081 [−0.093, −0.068]	−0.122 [−0.139, −0.105]	< 0.001	< 0.001
Num of judges	0.051 [−0.054, 0.157]	0.096 [−0.041, 0.234]	0.339	0.170
Model	−0.543 [−0.927, −0.159]	−0.703 [−1.059, −0.346]	0.006	< 0.001
TV personality	−0.438 [−0.648, −0.229]	−0.090 [−0.275, 0.095]	< 0.001	0.340
Athlete	−0.264 [−0.466, −0.061]	0.008 [−0.170, 0.186]	0.011	0.930
Comedian	−0.537 [−0.983, −0.091]	−0.074 [−0.484, 0.336]	0.018	0.725
Singer/Rapper	−0.023 [−0.238, 0.193]	−0.058 [−0.249, 0.133]	0.835	0.551
Social Media	0.033 [−0.462, 0.527]	−0.246 [−0.674, 0.181]	0.897	0.258
Other	−0.311 [−0.591, −0.031]	0.033 [−0.218, 0.284]	0.029	0.799

Report on Use of AI Tools

Team Statement (Human Decision Control). We used AI tools only to support idea exploration, code troubleshooting, and drafting. AI outputs were treated as hypotheses, not authority. Every suggestion was checked against our data, equations, and program outputs, and the team explicitly decided whether to accept, modify, or reject it. No model choice, parameter setting, or conclusion was adopted solely because the AI suggested it.

1. GPT-5.2thinking (Jan 30, 2026) — Official Dataset Audit and Cleaning Boundaries

Query: "Before modeling, how can we systematically audit the official dataset for missing values, inconsistent identifiers, duplicated records, and week-level anomalies, and define defensible cleaning rules without distorting show outcomes?"

Output:

- Run a structured audit for missingness (by column and by week), duplicated rows, and inconsistent keys (season/week/contestant identifiers).
- Flag rule-sensitive anomalies (e.g., no-elimination weeks, multi-elimination weeks, non-competing weeks) and treat them explicitly in downstream evaluation.
- Prefer minimal, reproducible cleaning: drop rows with missing `judge_total` or missing fan-share estimates rather than imputing unverifiable values.
- Keep a cleaning log that records each rule and the number of affected rows, so results can be reproduced and reviewed.

Application: We audited the provided dataset and implemented a minimal cleaning pipeline: we removed records with missing `judge_total` or missing estimated fan-share variables, standardized season/week identifiers, and explicitly excluded weeks with no elimination from counterfactual elimination comparisons. All changes were documented in a cleaning log for traceability.

Key decision: We adopted *minimal, auditable* cleaning and rejected imputation, because imputing missing judge totals or fan shares would introduce assumptions that cannot be validated from the official records.

2. Gemini 3.0 Pro (Jan 30, 2026) — Interval Estimation for Missing Fan-Vote Shares

Query: "Given that weekly fan-vote totals are unobserved, what defensible approaches can we use to infer *interval estimates* (a feasible range) for each contestant's fan-vote share using only the observed eliminations and weekly judge scores?"

Output:

- **Approach A (Constraint Set / Interval Inference):** Write the show's combination rule (Rank or Percent) and translate "the eliminated contestant has the worst combined result" into a system of inequalities over fan-vote shares $p_{i,t}$. Solve these constraints to obtain a feasible set (polytope) and per-contestant intervals; choose a representative point via maximum entropy or minimum-change criteria.

- **Approach B (Latent Vote Share with MLE/Bayes):** Treat $p_{i,t}$ as latent variables generated by a preference model (e.g., softmax of a utility $u_{i,t}$). Use eliminations (or bottom-two events) as observations to build a likelihood and estimate posteriors/intervals via regularized MLE or Bayesian inference.

- **Approach C (Survival/Hazard Modeling):** Model elimination as an event with hazard depending on judge scores and a latent popularity state; infer week-level popularity and map it to vote shares by normalization.

- **Approach D (Bradley–Terry / Pairwise Preferences):** Convert eliminations/bottom-two information into pairwise "less preferred" comparisons and infer relative popularity strengths; map strengths to shares if needed.

- **Approach E (State-Space Popularity):** Model popularity as a smooth latent state evolving over weeks; connect eliminations to the latent state via a probabilistic observation model and infer time-varying intervals.

Application: We adopted Approach A as the primary framework: for each week we formed inequality constraints implied by the elimination outcome and computed feasible intervals for $p_{i,t}$; we then produced a point estimate using a smooth, minimum-change selection rule within the feasible set.

Key decision: We prioritized a constraint-based feasible-set approach because it yields transparent interval uncertainty directly from the observed elimination logic, avoiding unverifiable assumptions required by more parametric latent-variable models.

3. GPT-5.2thinking (Jan 30, 2026) — Validating the Plausibility of Fan-Vote Intervals

Query: "After constructing feasible intervals for weekly fan-vote shares, how can we validate that these ranges are reasonable and not artifacts of overly-loose constraints?"

Output:

- **Elimination-consistency check:** Verify that there exists at least one feasible vote-share vector each week that reproduces the observed elimination under the assumed aggregation rule.

- **Boundary stress tests:** For each week, test whether extreme interval endpoints (lower/upper bounds) still allow feasibility; flag weeks where feasibility relies on degenerate corner cases.

- **Holdout validation:** Recompute intervals after removing one week (leave-one-week) or one contestant (leave-one-contestant) to assess stability of ranges.

- **Cross-rule robustness:** Compare interval widths under Rank vs Percent assumptions; large discrepancies indicate model sensitivity and should be reported.

- **Uncertainty diagnostics:** Summarize interval width and feasible-set dispersion (e.g., entropy from feasible samples) to identify weeks where fan votes are weakly identified.

Application: We validated weekly feasibility by explicitly checking constraint satisfaction and confirming at least one feasible solution per week. We then ran leave-one-week tests and recorded interval-width/entropy summaries to identify weeks with high uncertainty.

Key decision: We reported interval widths (and entropy) as primary uncertainty measures and treated weeks with unstable or corner-case feasibility as low-certainty, rather than forcing overly precise fan-vote estimates.

4. GPT-5.2thinking (Jan 31, 2026) — Cross-Applying Voting Rules Across Seasons

Query: "Should we apply *both* aggregation methods (Rank and Percentage) to *every* season—including seasons where the show used the other method—to generate a complete counterfactual dataset and compare outcomes against the observed results? If so, what is the most defensible comparison protocol?"

Output:

- Yes—cross-applying both rules to each season produces a clean, apples-to-apples comparison of rule behavior, avoiding confounding from season-to-season differences in scoring scales and voting variance.
- Use a **week-by-week counterfactual** protocol on the observed competitor set each week to avoid imputing unobserved future judge scores after an alternate elimination.
- For each week, compute the eliminated contestant under Rank and under Percent using the same inputs (judge totals and estimated fan shares), then record whether the predicted elimination matches the observed one.
- Aggregate results into season-level summaries: conflict-week rate, agreement with observed eliminations, and fan-alignment metrics (e.g., Spearman correlation between combined score and fan share).

Application: We generated a unified cross-season table by applying both Rank and Percent to every week in every season (using the same estimated fan shares). We then compared (i) elimination agreement with the observed outcome and (ii) conflict weeks where the two rules disagree.

Key decision: We adopted cross-application with week-by-week counterfactual evaluation as the primary comparison, because it isolates the effect of the aggregation rule without requiring speculative season-long trajectory imputation.

5. Gemini 3.0 Pro (Jan 31, 2026) — Controversy Cases and Judge-Save Counterfactuals

Query: "For controversial celebrities with large judge–fan disagreement, would using Rank vs. Percentage aggregation lead to the same elimination outcomes? Moreover, how would an additional *Judge-Save* rule (judges choose which couple to eliminate from the bottom two each week) change the results?"

Output:

- Identify **controversy weeks** by large judge–fan disagreement (e.g., large rank gap or large score–share mismatch) and focus counterfactual analysis on those weeks.

- Run a **week-by-week parallel-universe simulation**: for each week compute eliminations under (i) Rank, (ii) Percent, and (iii) Judge-Save (deterministically eliminating the bottom-two couple with the worse judge rank/score).
- Compare each method by (a) elimination week of the controversial contestant, (b) frequency of method-induced changes (ChangeRate), and (c) whether the contestant is systematically protected or penalized under each rule.
- Use the same observed weekly competitor set to avoid imputing future judge scores after alternative eliminations; report differences as counterfactual *week-level* changes.

Application: We evaluated known controversy examples (e.g., Jerry Rice, Billy Ray Cyrus, Bristol Palin, Bobby Bones) by computing, for each week they competed, the elimination under Rank, Percent, and Judge-Save. We summarized whether the three methods agree and how often Judge-Save overrides fan-driven eliminations.

Key decision: We adopted a three-method counterfactual protocol (Rank vs. Percent vs. Judge-Save) on controversy weeks to isolate rule effects under judge–fan disagreement, instead of relying on qualitative claims about "robbery" or "fan bias."

6. GPT-5.2thinking (Jan 31, 2026) — Modeling Pro-Dancer Effects and Celebrity Characteristics

Query: "Using our estimated fan-vote shares together with judge scores, what modeling approach can quantify (i) professional dancer (pro) effects and (ii) celebrity characteristics (e.g., age, industry), and do these factors affect judges and fans in the same way?"

Output:

- Use a mixed-effects (hierarchical) model to separate population-level drivers (fixed effects) from persistent unobserved heterogeneity (random effects).
- Include fixed covariates such as standardized age, industry categories, week index, and number of judges; optionally include season fixed effects for comparability.
- Add a random intercept for *pros* to capture partner-specific net contributions, and a separate random effect component for *celebrities* to capture person-specific baseline popularity/skill not explained by observed traits.
- Construct outcomes that are comparable across weeks: standardize judge totals within (season, week), and define the fan outcome as log relative vote share to remove mechanical scaling as the field size shrinks.
- Validate robustness by refitting with alternative outcome constructions (e.g., alternative vote-share estimates or alternative score standardization) and checking sign/stability of key coefficients.
- Test generalization with season-blocked validation and report both predictive error (RMSE) and a rule-consistency diagnostic (e.g., whether predictions place eliminated contestants in the bottom set).

Application: We fit two MixedLM specifications, one for judges and one for fans, using the same covariate set for interpretability. We defined `judge_z` by within-(season,week) standardization of judge totals and defined the fan outcome as log(relative vote share) based on the estimated fan-vote shares. We included a pro random intercept and a celebrity random component to isolate partner effects from contestant traits, and we evaluated season-blocked performance using RMSE and elimination-consistency checks.

Key decision: We adopted mixed-effects modeling with carefully normalized weekly outcomes (instead of pooled regression on raw scores/shares) so that pro effects and celebrity traits are identifiable, comparable across weeks, and testable for cross-season stability.

7. Gemini 3.0 Pro (Feb 1, 2026) — Measuring Effect Size and Judge–Fan Differences

Query: "How should we quantify how much pro-dancer effects and celebrity characteristics impact competition performance, and do these factors influence judges' scores and fan votes in the same way?"

Output:

- **Use matched outcomes for judges vs. fans:** Fit two parallel models (judge outcome and fan outcome) with the same covariates so coefficient differences are interpretable.
- **Report effect sizes, not only significance:** For fixed effects, report standardized coefficients (or marginal effects) with confidence intervals; for pro effects, report random-effect estimates with uncertainty and rank pros only when sample sizes are adequate.
- **Explain variance decomposition:** Use mixed-model variance components (and marginal/-conditional R^2) to quantify how much variation is explained by traits vs. pro/celebrity random effects.
- **Judge–fan comparison tests:** Compare coefficient signs and magnitudes across the judge and fan models; optionally test differences via bootstrapping or by fitting a combined model with an interaction term (OutcomeType × Covariate).
- **Practical impact summaries:** Translate coefficients into interpretable statements (e.g., "one SD increase in age corresponds to X SD change in judge_z"), and compare these statements between judges and fans.

Application: We fit MixedLM models for standardized judge totals and log relative fan-share using the same covariate set. We reported standardized fixed-effect coefficients with uncertainty, extracted variance components and marginal/conditional R^2, and compared coefficient directions and magnitudes to determine whether traits/pro effects align or diverge between judges and fans.

Key decision: We evaluated impact using effect-size and variance-explained metrics (not p-values alone) and used parallel judge/fan models to directly assess whether the same factors influence the two voting systems differently.

8. GPT-5.2thinking (Feb 1, 2026) — Dynamic Intensity and Out-of-Sample Generalization

Query: "How should rule strength vary across weeks, and how can we avoid overfitting to a small number of controversial seasons when tuning parameters?"

Output:

- **Scale with field size for interpretability:** Let the intervention strength depend on the number of active contestants N_t (e.g., $\lambda_t = \beta N_t$) so early weeks (large fields) allow modest protection while late weeks (small fields) naturally reduce intervention.
- **Avoid ad-hoc week tuning:** Use a fixed functional form for λ_t across all weeks to prevent "hand-tuning" around specific eliminations.
- **Temporal validation to prevent leakage:** Tune hyperparameters on earlier seasons and evaluate on later seasons to reflect real-world deployment and to avoid using future-season patterns to justify past rules.
- **Multi-metric selection:** Choose parameters by jointly considering fairness and fan agency (e.g., maximize RescueRate while keeping balance gap $\Delta\rho$ small and ChangeRate moderate), rather than optimizing a single headline statistic.
- **Stress-test sensitivity:** Confirm that the selected parameters remain competitive across nearby grid values (a "flat optimum"), indicating robustness rather than a fragile, overfit choice.

Application: We adopted the linear schedule $\lambda_t = \beta N_t$ and performed a temporal split: Seasons 1–17 were used for grid search over (α, β, p), and Seasons 18–34 were held out for out-of-sample evaluation. We reported RescueRate, $\Delta\rho$, ChangeRate (and fan-impact diagnostics) on both splits to verify that performance generalizes beyond the training era.

Key decision: We kept the transparent linear schedule and temporal train/test split because they are reproducible and reduce overfitting risk; we rejected ad-hoc week-specific tuning and parameter choices that improved training metrics at the expense of test stability.

9. Gemini 3.0 Pro (Feb 1, 2026) — Code-Level Sanity Checks and Tie Handling

Query: "What implementation pitfalls matter most for rank-based elimination, and how should ties be handled to keep results correct and reproducible?"

Output:

- **Directionality clarity:** Make the elimination direction explicit and consistent across scoring conventions (e.g., eliminate *argmax* for risk-style scores, but eliminate *argmin* for "higher-is-better" safety scores).
- **Ranking conventions:** Define rank order (1 = best), specify how ties are ranked (average rank), and ensure ranks are computed on the correct quantity (raw judge totals vs. normalized shares).
- **Deterministic tie-breaks:** Use a fixed tie-break order aligned with the rule intent (e.g., judge rank $\rightarrow$ fan rank $\rightarrow$ raw judge score $\rightarrow$ fan share) to ensure reproducibility.
- **Edge cases:** Handle multi-elimination weeks (eliminate top-k by risk), no-elimination weeks (skip or record), and non-standard exits (withdrawals) without silently corrupting weekly indices.

- **Invariants and sanity assertions:** Check that vote shares sum to 1 each week, all ranks lie in $\{1, \ldots, N_t\}$, and computed eliminations belong to the active contestant set.
- **Spot-checking:** Hand-calculate representative weeks (including a tie week if present) to verify that the code reproduces intended logic and does not invert comparisons.

Application: We implemented average-rank tie handling and a deterministic tie-break order, explicitly separated "risk-score" eliminations (argmax) from "percent-score" eliminations (argmin), and added basic assertions for weekly invariants (valid ranks, active-set membership, and normalized shares). We then hand-checked representative weeks to confirm directionality and eliminate inverted-logic bugs.

Key decision: We kept deterministic tie-breaking and explicit invariants because they guarantee reproducible competition reporting; random tie resolution and implicit ranking conventions were rejected as error-prone.

10. GPT-5.2thinking (Feb 1, 2026) — Model Selection Alternatives and Benchmarking Discipline

Query: "Which aggregation rules should be compared before recommending our final system, and how can we benchmark them fairly without cherry-picking?"

Output:

- **Define a clear candidate set:** Compare established baselines (Rank and Percent) against simple interpretable alternatives (weighted average, Borda/rank-fusion), and our proposed rule (MARS-style gated protection).
- **Use a single evaluation protocol:** Apply each rule under the same week-by-week counterfactual setup on the observed competitor set to avoid trajectory-imputation assumptions.
- **Multi-criteria decision:** Evaluate each rule along multiple axes—fairness (RescueRate), balance between judges and fans ($\Delta\rho$), fan agency (FII), and outcome disruption (ChangeRate).
- **Reject "degenerate" winners:** Discard rules that trivially optimize one metric by collapsing another (e.g., a rule that maximizes RescueRate by effectively ignoring fan votes).
- **Robustness checks:** Ensure the conclusion is not sensitive to small parameter perturbations or a single controversial season (e.g., verify performance on a temporal holdout split).

Application: We benchmarked the proposed MARS rule against Rank and Percent using the same week-by-week counterfactual protocol and reported key metrics side-by-side. We additionally used a temporal split to confirm that improvements persist out of sample rather than being driven by a few high-profile controversy cases.

Key decision: We recommended MARS only after it improved fairness (higher RescueRate) while maintaining meaningful fan influence (nontrivial FII and small $\Delta\rho$). Weighted-average variants were rejected when they amplified fan-noise sensitivity or produced unstable behavior on holdout seasons.

Tools Used (Disclosure). Gemini 3.0 Pro and GPT-5.2thinking, accessed on Jan 30, 2026, Jan 31, 2026, and Feb 1, 2026. No automated translation tools were used for the final report text. All analyses, model formulations, parameter selections, and conclusions were finalized by the team after independent verification.